DBMS Assignment-3

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Section :- 'A'

Section :- A

1. What is SQL? Why important?

Ans :- SQL (Structured Query language) is used to create, manage & retrieve data from relational database. It is important because it allows efficient data storage, querying & manipulation in DBMS.

2. Difference between DDL & DML.

Ans :-

DDL	DML
Define structure	Manipulate data
CREATE, ALTER, DROP.	INSERT, UPDATE, DELETE
Affect schema	Affect records.

→ DDL change database structure, while DML works with the actual data.

3. What is JOIN? Explain its type.?

Ans :- Join is used to combine data from multiple tables based on a related column.

Types:

- INNER JOIN → only matching rows
- LEFT JOIN → all left → matching right
- RIGHT JOIN → all right → matched left.
- FULL JOIN → all records.

4. What is GROUP BY and HAVING clause?

Ans GROUP BY → groups rows into categories.

HAVING ↳ filters grouped results.

Ex ↳ idea "count students per departements".

5. What is a subquery?

Ans: A subquery is a query inside another query, used to perform complex operation.

Section B

1 Write an SQL query to retrieve all student's with marks > 75.

Ans SQL
SELECT * FROM students
WHERE marks >75.

2. Write a query using JOIN to combine Students & Course tables.

Section-A

Ans 2 SQL

```
SELECT students. name, courses. course_name.
FROM students.
JOIN courses.
ON students. course_id = courses. id;
```

3. Write a query using GROUP BY to count students per department.

Ans: SQL

```
SELECT department, COUNT(*). AS total- students.
FROM students.
GROUP BY Department.
```

SECTION= C

1. A company database stores employee & department data. Write a query to find employee in a specific department.

Ans SQL

```
SELECT e.name.
FROM employees.
JOIN departments
ONe. dept_id = d.id.
WHERE d. name = 'HR';
```

2. An e-commerce system stores orders. Retrieve total sales per customer.

Ans SQL

```
SELECT customers_id, SUM (amount) As total_sales
FROM orders.
GROUP BY customer_id;
```

3 A university database needs top 3 student. Write a query.

Ay. SQL

```
SELECT name, marks
FROM students.
ORDER BY Marks DESC
LIMIT 3;
```

SECTION: D

1. A company database stores employee & department data write a query to find employee in a specific department.

Ay SQL

```
SELECT product_name, SUM (quantity) As total_sold
FROM Sales.
GROUP BY Product_name;
```

2. A university database needs to top 3 students. Write a query.

Ay: SQL

```
SELECT orders.id, customers.name
FROM orders.
JOIN customers.
ON orders.customer_id = customers.id;
```